

THE DEGREE-SIXTEEN COHOMOLOGY OF $\overline{\mathcal{M}}_{6,6}$ IS TATE

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ABSTRACT. We determine the Hodge type of the degree-sixteen rational cohomology of the moduli stack of stable genus-six curves with six ordered markings. We prove that

$$H^{16}(\overline{\mathcal{M}}_{6,6}; \mathbf{Q}) \cong \mathbf{Q}(-8)^{\oplus b_{16}}$$

as a polarizable rational Hodge structure. The argument combines the boundary exact sequence of Canning–Larson–Payne with a calculation of its open and boundary terms. For the open term, their description of the primitive quotient of lowest-weight cohomology, together with Harer’s virtual cohomological dimension, reduces the calculation to the inclusive endpoint of the known range for the Chow–Künneth generation property. For the boundary term, we prove by induction that $H^{16}(\overline{\mathcal{M}}_{g,n}; \mathbf{Q})$ is Tate for $g \leq 5$, $2g+n \leq 18$, and also for $g = 6$, $n \leq 5$. This includes the nonseparating boundary atlas $\overline{\mathcal{M}}_{5,8}$; all separating boundary atlases are handled by the Künneth formula, the vanishing of odd cohomology through degree nine, and the known Tate result in even degrees at most fourteen. The proof determines Hodge type but does not compute b_{16} and does not assert that the compact cohomology is generated by algebraic or tautological classes.

1. INTRODUCTION

Let $\overline{\mathcal{M}}_{g,n}$ denote the smooth proper Deligne–Mumford stack of stable complex curves of genus g with n ordered marked points, and let $\mathcal{M}_{g,n} \subset \overline{\mathcal{M}}_{g,n}$ be its open substack of smooth pointed curves. All cohomology and homology groups in this paper have rational coefficients. They are considered together with their natural mixed Hodge structures.

The rational cohomology of $\overline{\mathcal{M}}_{g,n}$ is known to be Tate in every even degree at most fourteen, uniformly in g and n [5, Theorem 1.5(3) and the discussion following it]. In a recent survey, Payne records that the corresponding general predictions in degrees 16, 18, and 20 remain open [10, Section 4]. The purpose of this paper is to establish a degree-sixteen case at the first of these open degrees.

Date: 1 September 2026.

Our main theorem is the following. The integer b_{16} in its statement is the sixteenth Betti number; no formula for it is asserted.

Theorem 1.1. *Let*

$$b_{16} := \dim_{\mathbf{Q}} H^{16}(\overline{\mathcal{M}}_{6,6}; \mathbf{Q}).$$

There is an isomorphism of polarizable rational Hodge structures

$$H^{16}(\overline{\mathcal{M}}_{6,6}; \mathbf{Q}) \cong \mathbf{Q}(-8)^{\oplus b_{16}}.$$

The proof has two parts, corresponding to the two terms surrounding $H_{16}(\overline{\mathcal{M}}_{6,6})$ in the boundary exact sequence. The open part is a lowest-weight calculation on $\mathcal{M}_{6,6}$. In fact, we obtain the following stronger description, in which RH^* denotes tautological cohomology.

Theorem 1.2. *For some nonnegative integer r ,*

$$\begin{aligned} W_{26}H^{26}(\mathcal{M}_{6,6}; \mathbf{Q}) &= \mathrm{RH}^{26}(\mathcal{M}_{6,6}; \mathbf{Q}) \cong \mathbf{Q}(-13)^{\oplus r}, \\ W_{-16}H_{16}^{\mathrm{BM}}(\mathcal{M}_{6,6}; \mathbf{Q}) &\cong \mathbf{Q}(8)^{\oplus r}. \end{aligned}$$

The boundary of $\overline{\mathcal{M}}_{6,6}$ contains one nonseparating component whose ordered atlas is $\overline{\mathcal{M}}_{5,8}$, as well as separating components whose atlases are products of smaller moduli stacks. To control them without presupposing the desired $(6,6)$ result, we prove the following auxiliary theorem. A pair (g, n) is called stable if $2g - 2 + n > 0$.

Theorem 1.3. *Suppose that (g, n) is stable and satisfies*

$$g \leq 5 \quad \text{and} \quad 2g + n \leq 18, \quad \text{or} \quad g = 6 \quad \text{and} \quad n \leq 5. \quad (1)$$

Then, for $b_{g,n} = \dim_{\mathbf{Q}} H^{16}(\overline{\mathcal{M}}_{g,n}; \mathbf{Q})$,

$$H^{16}(\overline{\mathcal{M}}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(-8)^{\oplus b_{g,n}}$$

as a polarizable rational Hodge structure.

The open input for Theorem 1.3 is already supplied by the published Chow–Künneth generation results except at $(g, n) = (3, 12)$ and $(5, 8)$. These two pairs and $(6, 6)$ share the same numerical feature. If $k = 2 \dim \mathcal{M}_{g,n} - 16$, then

$$k = \mathrm{vcd}(\mathcal{M}_{g,n}) \quad \text{and} \quad n - 1 = c(g),$$

where $c(g)$ is the marking bound in Table 1 of Canning–Larson–Payne [5]. Their primitive-quotient formula peels off one marking. All local-system terms and unmultiplied pullback terms then vanish in degrees strictly above the relevant virtual cohomological dimensions, while the remaining term is a cotangent-line class multiplied by lowest-weight cohomology at the inclusive endpoint $n - 1 = c(g)$. This proves that all three open inputs have the required Tate type.

The proof of Theorem 1.3 then follows the boundary induction of Canning–Larson–Payne, but it is useful to give the complete argument. On a product boundary atlas, an odd–odd Künneth summand in total degree 16 has one factor in odd degree at most seven and therefore vanishes by Bergström–Faber–Payne [2]. A positive even–even summand has both degrees at most fourteen and is Tate by [5, Theorem 1.5(3)]. The two axis summands are supplied by induction. The range in (1) is closed under passage to the vertex types of a one-edge stable graph, so the induction terminates.

Finally, Theorem 1.3 shows that the degree-sixteen cohomology of every component of the normalized boundary of $\overline{\mathcal{M}}_{6,6}$ is Tate. Hence the image of boundary homology is a quotient of a direct sum of copies of $\mathbf{Q}(8)$. The cokernel is the open group in Theorem 1.2. This makes $H_{16}(\overline{\mathcal{M}}_{6,6})$ an extension of Tate Hodge structures of the same weight; duality gives Theorem 1.1.

The conclusion concerns Hodge type only. A pure rational Hodge structure of type $(8, 8)$ need not be known to be generated by algebraic classes merely from its Hodge type. Accordingly, we do not claim that $H^{16}(\overline{\mathcal{M}}_{6,6})$ is tautological, nor do we make an integral statement or a statement about coarse moduli spaces.

2. HODGE-THEORETIC CONVENTIONS AND THE BOUNDARY SEQUENCE

2.1. Weights, twists, and duality. The Tate Hodge structure $\mathbf{Q}(m)$ has weight $-2m$ and Hodge type $(-m, -m)$. Thus $\mathbf{Q}(-8)$ has weight 16 and type $(8, 8)$, while $\mathbf{Q}(8)$ has weight -16 and type $(-8, -8)$. We use the increasing weight filtration $W_\bullet$. The mixed Hodge theory used below is due to Deligne [6, 7]; cohomology and duality for Deligne–Mumford stacks are discussed, for example, by Behrend [1].

If X is smooth of pure complex dimension d , Poincaré duality gives

$$H_i^{\text{BM}}(X; \mathbf{Q}) \cong H^{2d-i}(X; \mathbf{Q})(d) \quad (2)$$

as mixed Hodge structures. If X is proper, ordinary homology is the Hodge dual of ordinary cohomology:

$$H_i(X; \mathbf{Q}) \cong H^i(X; \mathbf{Q})^\vee. \quad (3)$$

For a smooth proper Deligne–Mumford stack, H^i is polarizable and pure of weight i , so H_i is polarizable and pure of weight $-i$.

We repeatedly use two elementary consequences. First, every subobject and every quotient of $\mathbf{Q}(m)^{\oplus N}$ in the category of rational mixed Hodge structures is again a finite direct sum of copies of $\mathbf{Q}(m)$. This follows directly from strictness of the Hodge and weight filtrations.

Second, the category of polarizable pure rational Hodge structures of a fixed weight is semisimple. Hence an extension of finite direct sums of a fixed Tate Hodge structure by finite direct sums of the same Tate Hodge structure splits noncanonically.

2.2. The normalized boundary. Write $\partial\widetilde{\mathcal{M}}_{g,n}$ for the normalization of the reduced boundary $\overline{\mathcal{M}}_{g,n} \setminus \mathcal{M}_{g,n}$, and let

$$\nu: \partial\widetilde{\mathcal{M}}_{g,n} \longrightarrow \overline{\mathcal{M}}_{g,n}$$

be its natural finite map. Knudsen's clutching constructions [9] give a decomposition

$$\partial\widetilde{\mathcal{M}}_{g,n} \cong \coprod_{\substack{[\Gamma] \text{ stable } \Gamma \text{ of type } (g,n) \\ |E(\Gamma)|=1}} [\overline{\mathcal{M}}_{\Gamma} / \text{Aut}(\Gamma)], \quad \overline{\mathcal{M}}_{\Gamma} := \prod_{v \in V(\Gamma)} \overline{\mathcal{M}}_{g_v, n_v}. \quad (4)$$

This is also the description used in [5, Section 2.1]. One may see the normalization directly from the local deformation space of a nodal curve: if t_e is the smoothing parameter of a node e , then the boundary is locally given by $\prod_e t_e = 0$, and its normalization records the choice of one node. Partial normalization at that distinguished node gives the one-edge graph in (4).

For a finite group G acting on a complex Deligne–Mumford stack Y , the Cartan–Leray spectral sequence and rational averaging give

$$H^*([Y/G]; \mathbf{Q}) \cong H^*(Y; \mathbf{Q})^G. \quad (5)$$

Indeed, $H^p(G, V) = 0$ for $p > 0$ over $\mathbf{Q}$. In particular, taking graph automorphism invariants cannot introduce a new Hodge type.

2.3. The right-exact boundary sequence. The localization sequence in Borel–Moore homology, together with strictness for weights, yields the following fundamental input.

Proposition 2.1 (Canning–Larson–Payne). *For every stable pair (g, n) and every integer i , there is a right-exact sequence of mixed Hodge structures*

$$H_i(\partial\widetilde{\mathcal{M}}_{g,n}; \mathbf{Q}) \xrightarrow{\nu_*} H_i(\overline{\mathcal{M}}_{g,n}; \mathbf{Q}) \longrightarrow W_{-i} H_i^{\text{BM}}(\mathcal{M}_{g,n}; \mathbf{Q}) \longrightarrow 0. \quad (6)$$

Proof. This is Equation (6.1) of [5]. Notice that the first map is ordinary proper pushforward in degree i . There is no degree shift and no Tate twist in (6). $\square$

At $i = 16$, let $I_{g,n}$ be the image of the first map. Then (6) gives a short exact sequence

$$0 \longrightarrow I_{g,n} \longrightarrow H_{16}(\overline{\mathcal{M}}_{g,n}; \mathbf{Q}) \longrightarrow W_{-16} H_{16}^{\text{BM}}(\mathcal{M}_{g,n}; \mathbf{Q}) \longrightarrow 0. \quad (7)$$

Thus the degree-sixteen problem separates into the Hodge type of a boundary image and the Hodge type of one precise lowest-weight group on the open moduli stack.

3. THE THREE ENDPOINT CALCULATIONS ON THE OPEN MODULI STACKS

3.1. Forgetful and cotangent-line subspaces. For $1 \leq i \leq n$, let

$$\pi_i: \mathcal{M}_{g,n} \longrightarrow \mathcal{M}_{g,n-1}$$

forget the i th marking, and let $\psi_i \in H^2(\mathcal{M}_{g,n}; \mathbf{Q})$ be the first Chern class of the corresponding cotangent line. Following [5, Sections 2 and 3], define

$$\Phi_{g,n}^k := \sum_{i=1}^n \pi_i^* W_k H^k(\mathcal{M}_{g,n-1}; \mathbf{Q}), \quad (8)$$

$$\Psi_{g,n}^k := \sum_{i=1}^n \psi_i \smile \Phi_{g,n}^{k-2}. \quad (9)$$

The first subspace consists of lowest-weight classes inherited from one fewer marking. The second is generated by multiplying such inherited classes by one cotangent-line class.

Let $\mathbb{V}_\lambda$ denote the irreducible symplectic local system on $\mathcal{M}_g$ indexed by a partition λ , and let V_{λ^τ} be the corresponding representation of the symmetric group. We use the following published description of the part not accounted for by (8) and (9).

Proposition 3.1 (Canning–Larson–Payne). *If $g \geq 2$, there is an isomorphism of pure rational Hodge structures*

$$\frac{W_k H^k(\mathcal{M}_{g,n}; \mathbf{Q})}{\Phi_{g,n}^k + \Psi_{g,n}^k} \cong \bigoplus_{|\lambda|=n} W_k H^{k-n}(\mathcal{M}_g; \mathbb{V}_\lambda) \otimes V_{\lambda^\tau}. \quad (10)$$

Proof. This is [5, Lemma 3.1(a)]. Its proof uses the smooth-proper Leray decomposition for powers of the closed universal curve. In particular, the identification of the index-two summands modulo the forgetful summands is made using the relative Chow–Künneth projectors described by Petersen–Tavakol–Yin [12, Sections 5.1 and 5.2.2]; it is not a consequence merely of restricting a cotangent-line class to a fibre. $\square$

3.2. Virtual cohomological dimension with rational coefficients.

We record the coefficient version of the virtual cohomological dimension vanishing that will be used in applying Proposition 3.1.

Lemma 3.2. *Let Γ be a virtually torsion-free group of virtual cohomological dimension d , and let L be a finite-dimensional rational Γ -module. Then*

$$H^q(\Gamma; L) = 0 \quad \text{for } q > d.$$

Proof. Choose a torsion-free normal subgroup $\Gamma' \triangleleft \Gamma$ of finite index and cohomological dimension d . The group $H^q(\Gamma'; L)$ vanishes for $q > d$. In the Hochschild–Serre spectral sequence for

$$1 \longrightarrow \Gamma' \longrightarrow \Gamma \longrightarrow F \longrightarrow 1,$$

the quotient F is finite. Taking F -invariants is exact over $\mathbf{Q}$, so $H^p(F, U) = 0$ for $p > 0$ and every rational F -module U . Consequently

$$H^q(\Gamma; L) \cong H^q(\Gamma'; L)^F = 0 \quad (q > d).$$

See [3, Chapter VII] for these standard group-cohomological facts. $\square$

Teichmüller space is contractible, so rational cohomology of $\mathcal{M}_g$ and $\mathcal{M}_{g,n}$, with the indicated local coefficients, agrees with the group cohomology of the mapping class groups Mod_g and $\text{PMod}_{g,n}$. Harer computed, for $g \geq 2$ and $n > 0$,

$$\text{vcd}(\text{Mod}_g) = 4g - 5, \quad \text{vcd}(\text{PMod}_{g,n}) = 4g - 4 + n \quad (11)$$

[8, Theorem 4.1].

3.3. The endpoint proposition. The numerical coincidences used in the proof are displayed in Table 1. Here $d = 3g - 3 + n$ and $c(g)$ is the bound in [5, Table 1].

TABLE 1. The three endpoint cases.

g	n	d	$k = 2d - 16 = 4g - 4 + n$	$c(g) = n - 1$
3	12	18	20	11
5	8	20	24	7
6	6	21	26	5

Proposition 3.3. *For each triple (g, n, k) in Table 1, there is a non-negative integer $r_{g,n}$ such that*

$$W_k H^k(\mathcal{M}_{g,n}; \mathbf{Q}) = \text{RH}^k(\mathcal{M}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(-k/2)^{\oplus r_{g,n}}. \quad (12)$$

Proof. Fix one of the three rows and apply Proposition 3.1. Since $k = 4g - 4 + n$, the cohomological degree on the right-hand side of (10) is

$$k - n = 4g - 4 > 4g - 5 = \text{vcd}(\text{Mod}_g).$$

Lemma 3.2 therefore makes every local-system summand vanish. Hence

$$W_k H^k(\mathcal{M}_{g,n}; \mathbf{Q}) = \Phi_{g,n}^k + \Psi_{g,n}^k. \quad (13)$$

The unmultiplied pullback term also vanishes. Indeed,

$$\mathrm{vcd}(\mathrm{PMod}_{g,n-1}) = 4g - 4 + (n - 1) = k - 1 < k,$$

so Lemma 3.2 gives $H^k(\mathcal{M}_{g,n-1}; \mathbf{Q}) = 0$. It follows from (8) that $\Phi_{g,n}^k = 0$, and therefore

$$W_k H^k(\mathcal{M}_{g,n}; \mathbf{Q}) = \Psi_{g,n}^k. \quad (14)$$

It remains to type the source of $\Psi_{g,n}^k$. In every row, $n - 1 = c(g)$. The bound is inclusive. By [5, Table 1, Lemma 4.3, and Proposition 4.5],

$$W_{k-2} H^{k-2}(\mathcal{M}_{g,n-1}; \mathbf{Q}) = \mathrm{RH}^{k-2}(\mathcal{M}_{g,n-1}; \mathbf{Q}).$$

This group is generated by codimension- $(k-2)/2$ algebraic cycle classes and is therefore a finite direct sum of copies of $\mathbf{Q}(-(k-2)/2)$. Each ψ_i is an algebraic class of type $\mathbf{Q}(-1)$. Cup product is a morphism of Hodge structures, so (9) and (14) show that $W_k H^k(\mathcal{M}_{g,n})$ is a quotient of a finite direct sum of copies of

$$\mathbf{Q}(-1) \otimes \mathbf{Q}(-(k-2)/2) = \mathbf{Q}(-k/2).$$

Thus it is itself a finite direct sum of $\mathbf{Q}(-k/2)$.

The same argument shows that every class in the group is tautological, so $W_k H^k \subseteq \mathrm{RH}^k$. Conversely, a degree- k tautological cycle class has Hodge type $(k/2, k/2)$ and weight k , hence belongs to $W_k H^k$. This proves the equality and the isomorphism in (12). All virtual-cohomological-dimension inequalities used above are strict; no vanishing is inferred from $k = \mathrm{vcd}(\mathcal{M}_{g,n})$. $\square$

We now translate Proposition 3.3 into the form required by the boundary sequence.

Corollary 3.4. *For each of $(g, n) = (3, 12), (5, 8), (6, 6)$,*

$$W_{-16} H_{16}^{\mathrm{BM}}(\mathcal{M}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(8)^{\oplus r_{g,n}}. \quad (15)$$

In particular, Theorem 1.2 holds with $r = r_{6,6}$.

Proof. For the three rows of Table 1, we have $k = 2d - 16$. Taking the lowest-weight piece in (2) gives

$$W_{-16} H_{16}^{\mathrm{BM}}(\mathcal{M}_{g,n}; \mathbf{Q}) \cong W_k H^k(\mathcal{M}_{g,n}; \mathbf{Q})(d).$$

By Proposition 3.3, the right-hand side is a direct sum of copies of $\mathbf{Q}(-k/2)(d)$. Since $k/2 = d - 8$,

$$\mathbf{Q}(-k/2)(d) = \mathbf{Q}(-(d-8) + d) = \mathbf{Q}(8).$$

Concretely, the three twist calculations are

$$\mathbf{Q}(-10)(18) = \mathbf{Q}(8), \quad \mathbf{Q}(-12)(20) = \mathbf{Q}(8), \quad \mathbf{Q}(-13)(21) = \mathbf{Q}(8).$$

The final case, together with (12), is exactly Theorem 1.2. $\square$

4. A TATE RANGE IN DEGREE SIXTEEN

We next prove Theorem 1.3. This section is independent of the compact (6, 6) conclusion: the pair (6, 6) itself does not lie in the range considered below.

4.1. Published low-degree inputs. We will use the following three results about compact moduli stacks.

Proposition 4.1. *For all stable pairs (g, n) :*

- (i) $H^j(\overline{\mathcal{M}}_{g,n}; \mathbf{Q}) = 0$ for odd $j \leq 9$;
- (ii) if $0 \leq j \leq 14$ is even, then $H^j(\overline{\mathcal{M}}_{g,n}; \mathbf{Q})$ is a finite direct sum of $\mathbf{Q}(-j/2)$;
- (iii) for every n , $H^{16}(\overline{\mathcal{M}}_{1,n}; \mathbf{Q})$ and $H^{16}(\overline{\mathcal{M}}_{2,n}; \mathbf{Q})$ are finite direct sums of $\mathbf{Q}(-8)$.

Proof. Part (i) is the odd-cohomology vanishing theorem of Bergström–Faber–Payne [2, Theorem 1.1]. Part (ii) follows from [5, Theorem 1.5(3) and the discussion following it]. For genus one, Petersen proves that even cohomology is generated by boundary-stratum cycle classes [11, Theorem 1.1]; in degree sixteen these have type (8, 8). For genus two, part (iii) follows from [5, Corollary 2.7], applied with $k = 8$. Purity of the compact cohomology turns these type statements into the displayed Tate decompositions. $\square$

The direct source of the open pure-weight calculations below is [5, Table 1 and Proposition 4.5]. Canning–Larson established the geometric results behind most entries of that table [4]. Here the table and proposition of [5] are the direct locators.

4.2. The numerical range and its boundary vertices. Let $\mathcal{R}$ be the set of stable pairs satisfying

$$\mathcal{R} := \{(g, n) : g \leq 5, 2g + n \leq 18\} \cup \{(6, n) : 0 \leq n \leq 5\}. \quad (16)$$

The following elementary identity is the reason this range is compatible with boundary induction.

Lemma 4.2. *Let Γ be a connected stable graph of type (g, n) . Then*

$$\sum_{v \in V(\Gamma)} (2g_v - 2 + n_v) = 2g - 2 + n. \quad (17)$$

Proof. Every leg occurs at one vertex and every edge contributes two half-edges, so $\sum_v n_v = n + 2|E|$. The arithmetic genus formula gives $\sum_v g_v = g - b_1(\Gamma)$, where $b_1(\Gamma) = |E| - |V| + 1$. Substitution gives

$$\begin{aligned} \sum_v (2g_v - 2 + n_v) &= 2(g - |E| + |V| - 1) - 2|V| + n + 2|E| \\ &= 2g - 2 + n. \end{aligned}$$

□

Lemma 4.3. *Let $(g, n) \in \mathcal{R}$, and let Γ be a one-edge stable graph of type (g, n) . Every vertex type (g_v, n_v) of Γ belongs to $\mathcal{R}$. Moreover, every factor $\overline{\mathcal{M}}_{g_v, n_v}$ has complex dimension strictly smaller than $\dim \overline{\mathcal{M}}_{g, n}$.*

Proof. Suppose first that $g \leq 5$ and $2g + n \leq 18$. Genus cannot increase on a vertex, so $g_v \leq 5$. Stability makes every summand on the left of (17) a positive integer. Hence

$$2g_v - 2 + n_v \leq 2g - 2 + n,$$

and therefore $2g_v + n_v \leq 2g + n \leq 18$. Thus every vertex lies in the first part of (16).

Now suppose $g = 6$ and $n \leq 5$. For the nonseparating graph the unique vertex has type $(5, n+2)$ and satisfies $2 \cdot 5 + (n+2) = 12 + n \leq 17$, so it lies in the first part of (16). For a separating graph, the two positive summands in (17) add to $10 + n \leq 15$, so each vertex satisfies $2g_v + n_v \leq 16$. A vertex of genus at most five lies in the first part of the range; a vertex of genus six then has at most four markings and lies in the second part.

Finally, partial normalization at one node lowers the total complex dimension by one. In the nonseparating case the single factor has dimension $3(g-1) - 3 + (n+2) = 3g - 4 + n$. In the separating case the factor dimensions add to

$$(3g_1 - 3 + n_1) + (3g_2 - 3 + n_2) = 3g - 4 + n.$$

Each individual factor therefore has dimension smaller than $3g - 3 + n$. □

4.3. The open term throughout the range.

Lemma 4.4. *Let $(g, n) \in \mathcal{R}$ with $g \in \{0, 3, 4, 5, 6\}$. Then*

$$W_{-16} H_{16}^{\text{BM}}(\mathcal{M}_{g, n}; \mathbf{Q}) \cong \mathbf{Q}(8)^{\oplus a_{g, n}}$$

for some $a_{g, n} \geq 0$.

Proof. Put $d = 3g - 3 + n$ and $j = 2d - 16$. If $j < 0$, the dual group in (2) is zero, and there is nothing to prove. Assume $j \geq 0$.

The marking bounds in [5, Table 1] are

$$c(0) = \infty, \quad c(3) = c(4) = 11, \quad c(5) = 7, \quad c(6) = 5.$$

The inequalities defining $\mathcal{R}$ give

g	0	3	4	5	6
largest possible n	18	12	10	8	5.

Thus $n \leq c(g)$ except possibly at $(3, 12)$ and $(5, 8)$. In the nonexceptional cases, [5, Proposition 4.5] gives

$$W_j H^j(\mathcal{M}_{g,n}; \mathbf{Q}) = \mathrm{RH}^j(\mathcal{M}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(-j/2)^{\oplus a_{g,n}}.$$

In the two exceptional cases the same conclusion is Proposition 3.3. Applying (2) and using $j = 2d - 16$ gives

$$W_{-16} H_{16}^{\mathrm{BM}}(\mathcal{M}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(-j/2)^{\oplus a_{g,n}}(d) = \mathbf{Q}(8)^{\oplus a_{g,n}},$$

as required. $\square$

4.4. Künneth decomposition in total degree sixteen.

Lemma 4.5. *Let (g_1, n_1) and (g_2, n_2) be stable pairs. Suppose, for $i = 1, 2$, that $H^{16}(\overline{\mathcal{M}}_{g_i, n_i}; \mathbf{Q})$ is a finite direct sum of $\mathbf{Q}(-8)$. Then*

$$H^{16}(\overline{\mathcal{M}}_{g_1, n_1} \times \overline{\mathcal{M}}_{g_2, n_2}; \mathbf{Q}) \cong \mathbf{Q}(-8)^{\oplus N}$$

for some $N \geq 0$.

Proof. The Künneth formula is an isomorphism of Hodge structures:

$$H^{16}(\overline{\mathcal{M}}_{g_1, n_1} \times \overline{\mathcal{M}}_{g_2, n_2}; \mathbf{Q}) \cong \bigoplus_{p+q=16} H^p(\overline{\mathcal{M}}_{g_1, n_1}; \mathbf{Q}) \otimes H^q(\overline{\mathcal{M}}_{g_2, n_2}; \mathbf{Q}).$$

If p and q are odd, then $\min(p, q) \leq 7$, so one factor vanishes by Proposition 4.1(i). If p and q are both positive and even, then $2 \leq p, q \leq 14$. Part (ii) of that proposition types the tensor product as a finite direct sum of

$$\mathbf{Q}(-p/2) \otimes \mathbf{Q}(-q/2) = \mathbf{Q}(-8).$$

The only remaining terms have $(p, q) = (16, 0)$ or $(0, 16)$. Since each moduli stack is connected, $H^0 \cong \mathbf{Q}(0)$, and the hypotheses type these axis terms as sums of $\mathbf{Q}(-8)$. $\square$

4.5. Induction on dimension.

Proof of Theorem 1.3. We prove the assertion simultaneously for all pairs in $\mathcal{R}$ by induction on $d = 3g - 3 + n$. If $d \leq 7$, then $16 > 2d$ and $H^{16}(\overline{\mathcal{M}}_{g,n}; \mathbf{Q}) = 0$. This is the base case.

Suppose $d \geq 8$. If $g = 1$ or $g = 2$, the conclusion is Proposition 4.1(iii), with no restriction on n . We may therefore assume $g \in \{0, 3, 4, 5, 6\}$. Lemma 4.4 gives

$$W_{-16}H_{16}^{\text{BM}}(\mathcal{M}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(8)^{\oplus a_{g,n}}. \quad (18)$$

Consider a component $[\overline{\mathcal{M}}_{\Gamma} / \text{Aut}(\Gamma)]$ of the normalized boundary. If Γ is nonseparating, its ordered atlas is a single factor $\overline{\mathcal{M}}_{g-1, n+2}$. By Lemma 4.3, this factor lies in $\mathcal{R}$ and has smaller dimension, so the induction hypothesis makes its degree-sixteen cohomology a direct sum of $\mathbf{Q}(-8)$.

If Γ is separating, its ordered atlas is a product of two moduli stacks. Both vertex types lie in $\mathcal{R}$ at smaller dimensions by Lemma 4.3. The induction hypothesis applies to the degree-sixteen cohomology of each factor, and Lemma 4.5 types the total degree-sixteen cohomology of the product. In either case, taking the finite graph-automorphism invariants in (5) preserves the type $\mathbf{Q}(-8)$. Summing over the finitely many one-edge graphs gives

$$H^{16}(\partial \widetilde{\mathcal{M}}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(-8)^{\oplus N_{g,n}}. \quad (19)$$

By proper duality (3),

$$H_{16}(\partial \widetilde{\mathcal{M}}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(8)^{\oplus N_{g,n}}. \quad (20)$$

Let $I_{g,n}$ be the boundary image in (7). It is a quotient of the group in (20), hence $I_{g,n} \cong \mathbf{Q}(8)^{\oplus e_{g,n}}$ for some $e_{g,n}$. Combining this with (18), the exact sequence (7) expresses $H_{16}(\overline{\mathcal{M}}_{g,n}; \mathbf{Q})$ as an extension of copies of $\mathbf{Q}(8)$ by copies of $\mathbf{Q}(8)$. This homology group is the dual of the polarizable pure weight-sixteen Hodge structure $H^{16}(\overline{\mathcal{M}}_{g,n}; \mathbf{Q})$, so it is polarizable and pure of weight -16 . Semisimplicity gives

$$H_{16}(\overline{\mathcal{M}}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(8)^{\oplus (a_{g,n} + e_{g,n})}.$$

Dualizing proves

$$H^{16}(\overline{\mathcal{M}}_{g,n}; \mathbf{Q}) \cong \mathbf{Q}(-8)^{\oplus b_{g,n}}.$$

The dimension has decreased at every use of the induction hypothesis, so the induction is well founded. This proves Theorem 1.3. $\square$

5. THE NORMALIZED BOUNDARY IN THE GENUS-SIX, SIX-POINTED CASE

We now specialize the graph description to the boundary of the stack in the main theorem. Its complex dimension is

$$\dim_{\mathbf{C}} \overline{\mathcal{M}}_{6,6} = 3 \cdot 6 - 3 + 6 = 21, \quad (21)$$

so every component of the normalized boundary has dimension twenty.

There is one nonseparating graph. Partial normalization at its loop lowers the genus by one and replaces the node by two markings. The two branches have no preferred order, and therefore the corresponding component is

$$[\overline{\mathcal{M}}_{5,8}/S_2], \quad (22)$$

where S_2 exchanges the last two markings.

For a separating one-edge graph, partial normalization produces two stable pointed curves with vertex types (g_1, n_1) and (g_2, n_2) satisfying

$$g_1 + g_2 = 6, \quad n_1 + n_2 = 8. \quad (23)$$

The two extra markings are the branches of the distinguished node. The component is the finite quotient

$$[(\overline{\mathcal{M}}_{g_1, n_1} \times \overline{\mathcal{M}}_{g_2, n_2}) / \text{Aut}(\Gamma)]. \quad (24)$$

Equations (22) and (24) exhaust the one-edge graphs.

Proposition 5.1. *There is a nonnegative integer N such that*

$$H^{16}(\partial \widetilde{\mathcal{M}}_{6,6}; \mathbf{Q}) \cong \mathbf{Q}(-8)^{\oplus N}, \quad (25)$$

$$H_{16}(\partial \widetilde{\mathcal{M}}_{6,6}; \mathbf{Q}) \cong \mathbf{Q}(8)^{\oplus N}. \quad (26)$$

Proof. The pair $(5, 8)$ satisfies $5 \leq 5$ and $2 \cdot 5 + 8 = 18$, so Theorem 1.3 gives

$$H^{16}(\overline{\mathcal{M}}_{5,8}; \mathbf{Q}) \cong \mathbf{Q}(-8)^{\oplus b_{5,8}}.$$

Taking S_2 -invariants therefore proves the required type for the nonseparating component (22).

Now consider a separating graph. Put $m_i = 2g_i + n_i$. From (23),

$$m_1 + m_2 = 2(g_1 + g_2) + (n_1 + n_2) = 20.$$

Stability gives $m_i \geq 3$, so $m_i \leq 17$. If $g_i \leq 5$, then (g_i, n_i) lies in the first part of the range (16). If $g_i = 6$, then $n_i = m_i - 12 \leq 5$, so it lies in the second part. Thus Theorem 1.3 applies to both factors of the ordered atlas. Lemma 4.5 and rational graph-automorphism invariants then give the type $\mathbf{Q}(-8)$ for every separating component.

The graph set is finite, so the direct sum over all components proves (25). Proper duality (3) gives (26). $\square$

The boundary map itself may have a nontrivial kernel. Its rank is not needed, because its source has only one Hodge type.

Corollary 5.2. *For some integer m with $0 \leq m \leq N$,*

$$\mathrm{im}\left(H_{16}(\partial\widetilde{\mathcal{M}}_{6,6}; \mathbf{Q}) \longrightarrow H_{16}(\overline{\mathcal{M}}_{6,6}; \mathbf{Q})\right) \cong \mathbf{Q}(8)^{\oplus m}.$$

Proof. The image is a quotient, in the category of mixed Hodge structures, of the source in (26). Every quotient of a finite direct sum of $\mathbf{Q}(8)$ is again a finite direct sum of $\mathbf{Q}(8)$. $\square$

It is worth noting that no cohomology group of $\overline{\mathcal{M}}_{6,6}$ was used to prove Corollary 5.2. The image is controlled entirely from the normalized boundary, so the argument does not feed the desired compact conclusion back into itself.

6. PROOF OF THE MAIN THEOREM

Proof of Theorem 1.1. Specialize (7) to $(g, n) = (6, 6)$. By Corollary 5.2, its boundary image is $\mathbf{Q}(8)^{\oplus m}$. By Corollary 3.4, its open quotient is $\mathbf{Q}(8)^{\oplus r}$. We therefore have a short exact sequence of mixed Hodge structures

$$0 \longrightarrow \mathbf{Q}(8)^{\oplus m} \longrightarrow H_{16}(\overline{\mathcal{M}}_{6,6}; \mathbf{Q}) \longrightarrow \mathbf{Q}(8)^{\oplus r} \longrightarrow 0. \quad (27)$$

The stack $\overline{\mathcal{M}}_{6,6}$ is smooth and proper, so the middle term is a polarizable pure Hodge structure of weight -16 . Semisimplicity splits (27) noncanonically and gives

$$H_{16}(\overline{\mathcal{M}}_{6,6}; \mathbf{Q}) \cong \mathbf{Q}(8)^{\oplus(m+r)}.$$

By proper duality,

$$H^{16}(\overline{\mathcal{M}}_{6,6}; \mathbf{Q}) \cong H_{16}(\overline{\mathcal{M}}_{6,6}; \mathbf{Q})^{\vee} \cong \mathbf{Q}(-8)^{\oplus(m+r)}.$$

The integer $m + r$ is the dimension of this cohomology group, hence equals b_{16} . This proves the theorem. $\square$

7. SCOPE OF THE RESULT

Remark 7.1 (Hodge type versus algebraic generation). Theorem 1.1 says that the complete degree-sixteen cohomology group has Hodge type $(8, 8)$. The argument does not exhibit a basis of cycle classes on $\overline{\mathcal{M}}_{6,6}$. Thus it does not prove that $H^{16}(\overline{\mathcal{M}}_{6,6})$ is tautological and does not, by Hodge type alone, prove surjectivity of the cycle class map in codimension eight.

Remark 7.2 (Ranks). The proof introduces three finite ranks: r for the open quotient, N for the normalized boundary, and m for the image of boundary homology. It shows $b_{16} = m + r$ and $0 \leq m \leq N$, but computes none of these integers. The theorem includes the possibility that any of the corresponding summands is zero.

Remark 7.3 (Coefficients and stacks). All statements are over $\mathbf{C}$ with rational coefficients and concern the cohomology of Deligne–Mumford stacks. Rational coefficients are used essentially when passing to invariants of finite graph automorphism groups. No integral assertion and no comparison with the cohomology of the coarse moduli space is made.

Remark 7.4 (Why the endpoint is sharp for this proof). The open calculation for $\mathcal{M}_{6,6}$ uses the equality $6 - 1 = c(6) = 5$; the calculations for $\mathcal{M}_{5,8}$ and $\mathcal{M}_{3,12}$ similarly use $8 - 1 = c(5) = 7$ and $12 - 1 = c(3) = 11$. The argument reaches one marking beyond the published Chow–Künneth range only because the primitive quotient and the unmultiplied pullback term vanish strictly above virtual cohomological dimension. No Chow–Künneth generation statement is asserted for the three endpoint stacks themselves.

ACKNOWLEDGMENTS

The principal proof development, mathematical analysis, drafting, and review underlying this manuscript were carried out by the artificial intelligence collaborators named in the byline.

REFERENCES

- [1] K. Behrend, *Cohomology of stacks*, in *Intersection theory and moduli*, ICTP Lecture Notes, vol. XIX, Abdus Salam International Centre for Theoretical Physics, Trieste, 2004, pp. 249–294.
- [2] J. Bergström, C. Faber, and S. Payne, *Polynomial point counts and odd cohomology vanishing on moduli spaces of stable curves*, *Ann. of Math. (2)* **199** (2024), no. 3, 1323–1365, [doi:10.4007/annals.2024.199.3.7](https://doi.org/10.4007/annals.2024.199.3.7).
- [3] K. S. Brown, *Cohomology of groups*, Graduate Texts in Mathematics, vol. 87, Springer-Verlag, New York, 1982, [doi:10.1007/978-1-4684-9327-6](https://doi.org/10.1007/978-1-4684-9327-6).
- [4] S. Canning and H. Larson, *On the Chow and cohomology rings of moduli spaces of stable curves*, *J. Eur. Math. Soc.* **28** (2026), no. 9, 3869–3918, [doi:10.4171/JEMS/1543](https://doi.org/10.4171/JEMS/1543).
- [5] S. Canning, H. Larson, and S. Payne, *Extensions of tautological rings and motivic structures in the cohomology of $\overline{\mathcal{M}}_{g,n}$* , *Forum Math. Pi* **12** (2024), article no. e23, 31 pp., [doi:10.1017/fmp.2024.24](https://doi.org/10.1017/fmp.2024.24).
- [6] P. Deligne, *Théorie de Hodge. II*, *Publ. Math. Inst. Hautes Études Sci.* **40** (1971), 5–57, [doi:10.1007/BF02684692](https://doi.org/10.1007/BF02684692).
- [7] P. Deligne, *Théorie de Hodge. III*, *Publ. Math. Inst. Hautes Études Sci.* **44** (1974), 5–77, [doi:10.1007/BF02685881](https://doi.org/10.1007/BF02685881).

- [8] J. L. Harer, *The virtual cohomological dimension of the mapping class group of an orientable surface*, Invent. Math. **84** (1986), no. 1, 157–176, doi:[10.1007/BF01388737](https://doi.org/10.1007/BF01388737).
- [9] F. F. Knudsen, *The projectivity of the moduli space of stable curves. II. The stacks $\overline{\mathcal{M}}_{g,n}$* , Math. Scand. **52** (1983), 161–199, doi:[10.7146/math.scand.a-12001](https://doi.org/10.7146/math.scand.a-12001).
- [10] S. Payne, *On the Hodge and Tate conjectures for moduli spaces of curves*, 2026, [arXiv:2605.20453v1](https://arxiv.org/abs/2605.20453v1).
- [11] D. Petersen, *The structure of the tautological ring in genus one*, Duke Math. J. **163** (2014), no. 4, 777–793, doi:[10.1215/00127094-2429916](https://doi.org/10.1215/00127094-2429916).
- [12] D. Petersen, M. Tavakol, and Q. Yin, *Tautological classes with twisted coefficients*, Ann. Sci. Éc. Norm. Supér. (4) **54** (2021), no. 5, 1179–1236, doi:[10.24033/asens.2479](https://doi.org/10.24033/asens.2479).